Probabilities and natural selection

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widows I heard wailing wildly
and children sobbing for bread —
Stars do not care even mildly
if someone is born or is dead.
(Nils Ferlin)

i

MUCH of what we notice around us are not things that are, but rather changes in what there is. We are desensitised to the common things, while the changes are something we need to notice and possibly evaluate. In philosophy, it is the other way around: Dynamicity is largely neglected by mainstream metaphysics (Seibt, 2025). All changes occur with some probability, or so we believe. In this note I try to describe what kinds of probabilities that lie behind processes in nature. In particular, I concentrate on the kind of probabilities we can trace in the theory of evolution by natural selection. There are problems connected with this goal; the most important one is, as pointed out by Bertrand Russell in a lecture 1929

> Probability is the most important concept in modern science, especially as nobody has the slightest notion what it means (cited from Hájek, 2023).

Little has changed during the century that has passed since Russell held that lecture. We have now sophisticated theories on probability, statistical inference and big data. I assume that all of them mention Kolmogorov's axioms, flipping coins and fair dice. However, I do not think any of these theories dispute Russell. Neither will I. However, in this note I argue that one *interpretation* of the concept has a better compatibility with the theory of evolution than the rest. Namely the propensity interpretation proposed by Popper (1959a).

This is an ontological claim. When modelling natural processes we need quantities, variables or constants, that we refer to as probabilities since they are about the likelihood that something happens out there. They are objective in the sense that they exist, whether or not we observe or estimate them. David Lewis writes a questionnaire for the readers of his paper 'A Subjectivist's Guide to Objective Chance' (Lewis, 1987). His first question is:

> A certain coin is scheduled to be tossed at noon today. You are sure that this chosen coin is fair: it has a 50% chance of falling heads and a 50% chance of falling tails. You have no other relevant information. Consider the proposition that the coin tossed at noon today falls heads. To what degree would you now believe that proposition?

He gives the answer '50%, of course.' I agree, could not do otherwise. Lin (2024) asks the same question and writes then

> According to the test, you are a Bayesian if your answer is "50%, because I am 50% sure that it landed heads, and equally sure that it didn't." And you are a frequentist if your answer is "the probability is unknown but equals either 1 or 0, depending on whether the coin actually landed heads or tails, because probabilities are frequencies of events."

So, I am a Bayesian; I feel a bit uneasy. Lin gives me a relief: '[...] this test is too simplistic to reveal the complexity underlying the seemingly binary question.' I have no problems with David Lewis' *Principal Principle* (Lewis, 1987; Lewis, 1994).

It is just that this note is not about that. I will try to show that natural selection is not operating through induction but falsification.

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We normally make a distinction between *subjective* and *objective* probabilities (Popper, 1957a). These are the two interpretations of probabilities. The subjective probability refers to an epistemic aspect of an event, for instance our confidence that a certain proposition about that event is true. Objective probabilities are those that exist in Nature no matter what. That is, we make metaphysical claims when using that concept. For instance, each of us carries ≈ 75.8 new mutations.[1] Mutation occurs all over the human genome at a rate from about 10^{-9} to 10^{-8} per base pair per generation. Still, the size of our genome $\approx 3.055\times10^{9}$ ensures that all of us are mutants. It has been like that since long before we became *Homo sapiens*. See Figure 1 for more information on mutations.

Popper makes further distinctions between two kinds of objective probabilities, namely frequency and propensity interpretations of probability (Popper, 1959a). The frequency interpretation claims that a probability can only be understood, and be estimated, through a series of measurements (possibly an infinitely long series).

A propensity should be understood as the probability for a single event (Popper, 1959a). Popper was convinced that probabilities are 'physically real',

> that they must be physical propensities, abstract relational properties of the physical situation, like Newtonian forces, and 'real', not only in the sense that they could influence the experimental results, but also in the sense that they could, under certain circumstances (coherence), interfere with one another (Popper, 1959a, p. 28).

That is, the key property of a propensity is that it behaves analogously to a force. Mutations are such forces. One opponent to Popper's idea of the assignment of probabilities to single events was von Mises (1957). von Mises talks about *classes* or *collectives.*

> When we speak of 'the probability of death', the exact meaning of this expression can be defined in the following way only. We must not think of an individual, but of a certain class as a whole, e.g., 'all insured men forty-one years old a given country and not engaged in certain dangerous occupations'. A probability of death is attached to this class of men or to another class that can be defined in a similar way (von Mises, 1957).

Given a defined class of objects we can estimate a meaningful probability. For instance

> The phrase 'probability of death', when it refers to a single person, has no meaning at all for us. This is one of the most important consequences of our definition of probability and we shall discuss this point in greater detail later on (von Mises, 1957, p. 11).

The collective means in von Mises' vocabulary basically an experimental series. The adherent of the propensity theory does not dispute that a series of measurements is needed for estimating the value of a probability. However, he or she regard the frequency approach as a method, and its interpretation as an epistemic problem. Popper admits readily that from the point of view of a 'purely statistical interpretation of probability' (Popper, 1959a, p. 30). the propensity interpretion is unacceptable. Still he maintains that it is useful.

> The concept of force—or better still, the concept of a field of forces—introduces a dispositional physical entity described by certain equations (rather than by metaphors), in observable accelerations. Similarly, the concept of propensity, or a field of propensities, introduces a dispositional property of singular physical experimental arrangements—that is to say, of events—in order to explain observable frequencies in sequences of repetitions of these events.

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[1] The mutation rate for humans is in the order of magnitude 1.24×10^{-8} per base pairs per generation (Tian *et al.*, 2022). Each of us carries around 3.055×10^{9} base pairs per chromosome set (Nurk et al, 2022) and we are diploid organisms (we have two sets of chromosomes, one from each parent). Therefore both you and I have got about 75.76 new mutations ($1.24\times10^{-8}\times3.055\times10^{9}\times2$) from our parents.

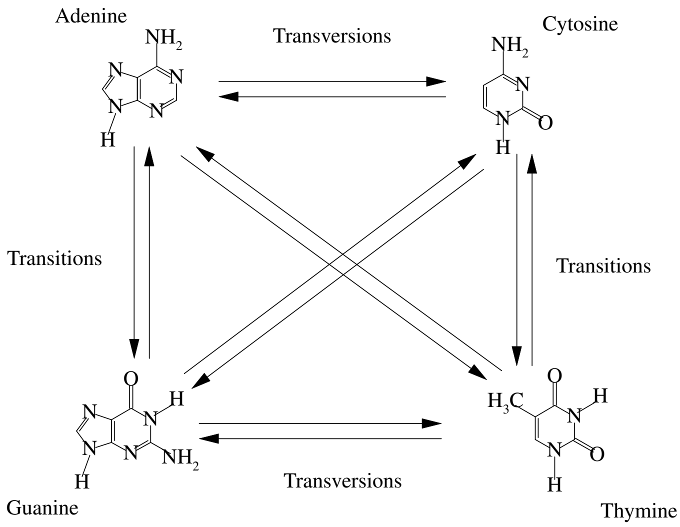

Figure 1. More than you ever wanted to know about mutations. There are two types of DNA substitution mutations. *Transitions* refer to substitutions between two-ring purines, i.e., an Adenine replaces Guanine (or vice versa), or a one-ring pyrimidine is replaced by another one-ring pyrimidine. That is Cytosine is replaced with a Thymine (or vice versa). One base is replaced with another with a similar shape. *Transversions* are more complicated interchanges of a purine for a pyrimidine, or vice versa. They imply exchanges one-ring and two-ring bases. The frequency of transitions is in the order magnitude 10^{-8} whereas transversions occur at a rate 10^{-9} per gene per generation (Nachman and Crowell, 2000; Tian *et al.*, 2022).

Let us return to the probability of death example raised by von Mises (1957). We are all mutants, and each of us will die. There is a connection between the deaths and the mutations, and that is *natural selection.* We start with mortality.

When discussing such probabilities we need a kind of data usually referred to as *life table* data. Those data are collected by researchers in various disciplines and are in high demand from insurance companies, health authorities and pension funds, just to name a few. They are important for many practical purposes, like calculating your insurance premium. In Figure 2 there are graphs showing the probability of living to a given age for Swedish women at three different periods in time, around 1800, 1900 and 2000, respectively. They are examples of a kind of graph called survivorship curves. See Rauschert (2010) for a popular description of survivorship curves, including curves for other organisms than humans.

The ordinate scale (on the y axis) is logarithmic, while the abscissa (x axis) is linear. There is a rapid fall at early ages (including infant mortality). Later the curves decline almost linearly till we reach perhaps 55–60 years of age when they bend downwards, the mortality accelerates. After that bend the decline is again almost linear, but very much more rapid. That final bend marks the onset of senescence. That the curves become linear on an ordinate logarithmic scale is significant.[2] When the curve is linear the yearly death rate is constant, which means that a small fixed proportion of the population of one age die *per annum.*

During such a period the constancy of the force of mortality is significant. It implies that the mortality rates are fairly constant during periods of life. The significance of this was pointed out by Peter Medawar (1957) who was one of the first to discuss ageing as an evolutionary problem. The problem is to give evolutionary explanations to the shapes of the survivorship curves. Genes that are favourable after the end of the reproductive phase of our lives will not spread in the next generation. Genes that enhance reproduction, or raises the survival rate up to and including the reproductive period, will have a chance to spread.

[2] Assume that at some moment in time we see the birth of N_0 infants. Then, if a proportion c of them dies per year, x years later $N_x = N_0 c^x$ are still alive. Taking the logarithm (ln) of that we see $\ln N_x = x \ln c + \ln N_0$, that is, a straight line with the slope $\ln c$.

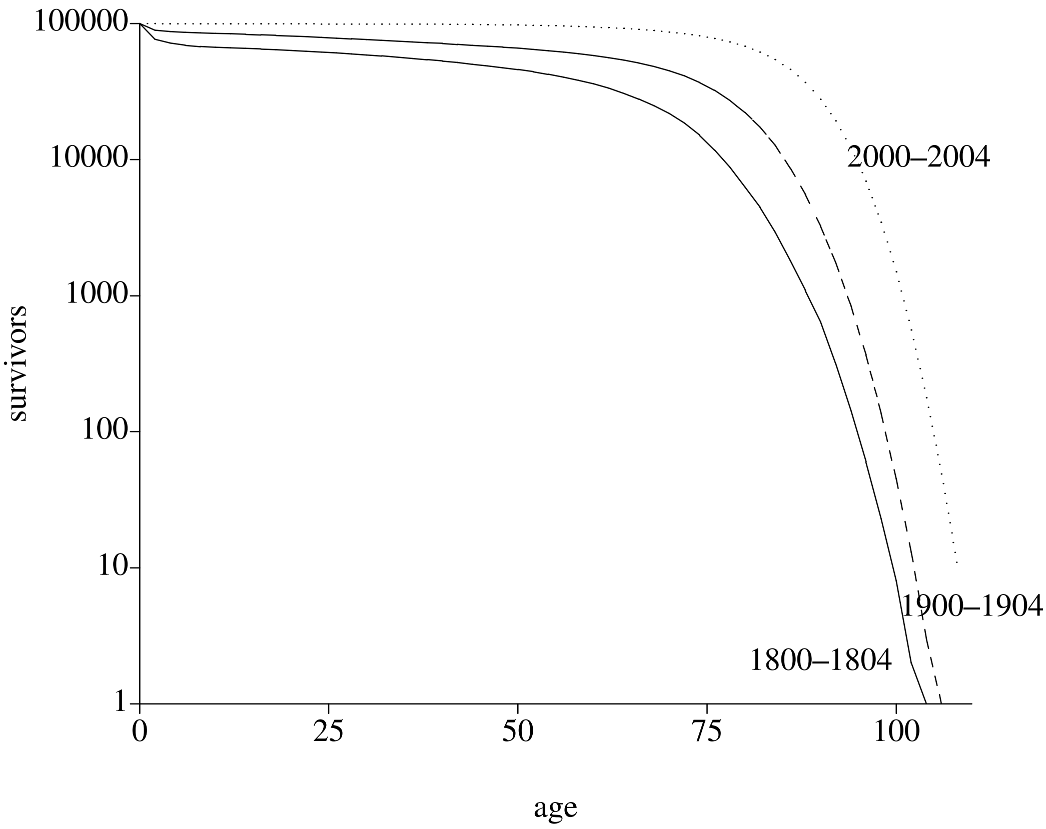

Figure 2. The survivorship curves for Swedish women during three five year intervals in history, namely, 1800-1804, 1900-1904 and 2000-2005 (solid, dashed and dotted curves, respectively). Data from Human Mortality Database (HMD: https://www.mortality.org/). Max Planck Institute for Demographic Research (Germany), University of California, Berkeley (USA), and French Institute for Demographic Studies (France). The data for the graph above are available at https://www.mortality.org/File/GetDocument/hmd.v6/SWE/STATS/fltper_1x5.txt (data downloaded on 2026-03-26)

However, there is no way a living being can improve in that way. Any improvement comes at a cost. Or as Williams puts it in his seminal paper:

> The selective value of a gene depends on how it affects the total reproductive probability. Selection of a gene that confers an advantage at one age and a disadvantage at another will depend not only on the magnitudes of the effects themselves, but also on the times of the effects. An advantage during the period of maximum reproductive probability would increase the total reproductive probability more than a proportionately similar disadvantage later on would decrease it. So natural selection will frequently maximize vigour in youth at the expense of vigour later on. It will therefore produce a declining health (senescence) during late adult life (Williams, 1957).

This is the gist of the socalled *pleiotropy* theory of senescence. Pleiotropy is the condition arising when one or a small group of genes have multiple phenotypic manifestations. The phenotype is jargon for what the genes do to their carriers. In this case the pleiotropy is stretching across time in a favourable way. You live through your youth and then you reproduce, but then you pay with arthritis, senility and stroke once your offspring has fledged.

For such an organism to evolve, in which individuals can be vigorous at for a time period long enough for allowing them to help their grandchildren requires a lot of mutations to generate variation needed, and a lot deaths and survivals to fine tune the balance between reproduction, senescence and mortality. Evolutionary gerontology is now a science in its own right (Li *et al.*, 2023).

Popper provides arguments for allowing singular events having probabilities based on sequence of throws of two dice. One die is homogenous and symmetrical and contribute a few throws, and the other is loaded and contribute a majority. Each throw becomes an individual event. So are the deaths of individuals—each of which is a mutant—in a population of other mutants. The additional feature is that these *individual* deaths contribute to the moulding of the survivorship curve for the population. The mutation pressure provides the variation that enables evolution, the selection pressure determines what changes occur in the population.

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All this raises the question how the theories of heredity and natural selection as we know them impact our understanding of probability. However, we must make two excursions into some theories. First, we need to take a look at how Karl Popper handles probability and, secondly, use that to analyse Frater Gregor Mendel's (OSA)[3] laws of heredity. Then I return to natural selection in a section of its own below.

Let us start with Popper. We have already mentioned his insistence on there being probabilities for singular events. Second, we note that he rejects explicitly the notion that probabilities exist outside a context. That is, all probabilities are conditional on something. This is not incompatible with mainstream probability theory, he is just making it a bit more general. The notion that all probabilities are conditional is tightly connected to his theory on falsification (Popper, 1957a; Popper, 1959b). Therefore any probability requires an interpretation, and by that he means that statements like

'The probability of a given b is equal to r (where $r \in \mathbb{R}$)'

implies, for example, that a can be a hypothesis which is studied in an experimental design b. That is, anyone using probabilities is obliged to explain what they are conditional on. He uses his own notation for this:

$$p(a, b) = r \tag{P1}$$

That probability (for a given b to be pedagogically redundant), can be written in a more conventional notation

$$P(a|b) = \frac{P(a \cap b)}{P(b)} \tag{P1'}$$

using the more common set theoretical notation rather than Popper's logical one. The two equations say the same things, with one exception. Eq. 1 is valid even in the case that $P(b) = 0$. Otherwise, the textbook notation $P(a|b)$ is what Popper calls r. In this note I will use Popper's notation and logic rather than set theory. Popper has defined his own calculus of probability containing three theorems. They are: The addition theorem, for calculating the probability of $a \vee b$ given c

$$p(a \vee b, c) = p(a, c) + p(b, c) - p(a \wedge b, c) \tag{P2}$$

This theorem does not assume that a and b are disjoint, Kolmogorov's first axiom does that. When that is the case, the event $a \wedge b$ given c will never occur and the third term in Eq. (P1) will be zero. See Figure 3 for a graphic interpretation of Eq. (P2). His multiplication theorem

$$p(a \wedge b, c) = p(a, b \wedge c)p(b, c) \tag{P3}$$

is really the same as

$$P(a \cap b|c) = P(a|b \cap c)P(b|c) \tag{P3'}$$

i.e., it comes from Eq. (P1'). You cannot see that from Eq. (P1), though.

[3] Frater Gregor belonged to *Ordo Fratrum Sancti Augustini*.
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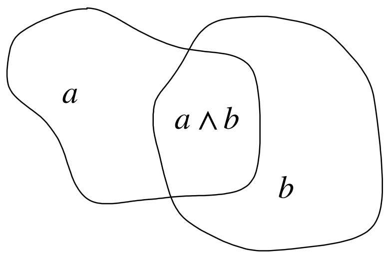

Figure 3. The two events a and b are represented as two sets. The intersection is the event $a \wedge b$ and the event $a \vee b$ must not double-count the overlapping area. Hence the third term in Eq. (P2). The event c is the whole background area. In a more modern notation this would be

$$P(a \cup b) = P(a) + P(b) - P(a \cap b)$$

Kolmogorov's third axion states

$$P(a \cup b) = P(a) + P(b)$$

and does not allow for the a and b overlapping as depicted.

Finally, let $\overline{a}$ be the complement of a. The complement of a set is everything which is not in that set.

$$p(\overline{a}, c) = 1 - p(a, c), \text{ provided } p(\overline{c}, c) \neq 0 \tag{P4}$$

Popper admits himself that Eq. (P4) is somewhat unusual.[4] I will not use it in this note, kept it here for the sake of completeness.

Let us return to Fra Gregor Mendel and his laws. In a series of experiments with crossings of peas he could demonstrate the principles of inheritance.[5] Mendel's most important observation was that both parents contribute equally to the genome of their offspring. Each individual contains two copies of the genome, one from each parent. In turn each individual passes on one such copy to each of their own offspring. A gene is the genetic contribution to a trait, usually called a phenotype after the environment (the nurture) has had its impact. He was the first to prove this. From (at least) Aristotle to Mendel the accepted truth was that the man's seed contained everything needed to make a little child.

We talk about *quantitative* and *qualitative* inheritance. In the former case there are a (possibly large) number of genes contributing to a single character. In the latter we have a single gene leading to qualitative characters, such as red and white pea flowers.

In the formation of germ cells the two parts of the genome separates randomly. That process is referred to as meiosis. (The division of cells in the body in general is called mitosis). For the male function it is formed sperms for animals as well as in pollen for flowering plants. Likewise, for the female function animal egg cells or plant ovules are formed. In the example worked on by Mendel we have red and white pea flowers: We have a red gene R which is *dominant* and a white one (W) which is not expressed at all when it is combined with a red. That is we have two phenotypes but three genotypes: RR and RW have both the phenotype red, and just the genotype WW is white. The white gene is *recessive* (Müntzing, 1971).

The place for a gene within the genome is called a *locus.* The loci are collected on *chromosomes.* Each gene that may occupy a locus is called an *allele.* A locus may have two

[4] This is because if c is true, i.e, a tautology, then its complement $\overline{c}$ has to be a contradiction. The probability of anything given something certain (a tautology) is 1 and Eq. (P4) would become

$$1 = 1 - 1$$

which is necessarily false. Requiring $p(\overline{c}, c) \neq 0$ precludes this.

[5] See Miko (2008) for a popular description of the achievements of Mendel.

$RW \times RW$

Pollen

Ovules	R	W
R	$\frac{1}{4}$ RR	$\frac{1}{4}$ RW
W	$\frac{1}{4}$ WR	$\frac{1}{4}$ WW

Figure 4. Diagram showing the crossing $RW \times RW$ studied by Mendel and discussed in the main text. The shaded area shows offspring with red pea flowers, the open square with white flowers. This is due to R being dominant, and W recessive.

alleles (for a diploid organism like me), as we saw above. An individual which is WW giving white pea flowers is referred to as a *homozygote,* whereas if we have two different alleles, RW, we have a *heterozygote*. Brown and blue eyes have in the simplest case similar inheritance. However, in general the colour of the iris depend on genes in at least three loci with several alleles in each of them (Wikipedia, 2026).

The Augustines themselves are justly proud of their Fra Gregor

> He spent two years preparing the material for his best-known experiment, the one with peas, until he was sure that he was working with pure breeds. He handled more than 27,000 plants and patiently crossed a large number of them, both a thankless and visually exhausting job. When he discovered the mathematical relationship between the inherited traits, thanks to the breadth of his experiment, he summarised it in a 45-page publication, giving an account of his nine years' work. Only scientific rigour and personal conviction that he was reaching important conclusions can explain his thoroughness, method and patience (Orcasitas, 2022).

The perhaps best known experiment in Mendel's study was his crosses between $RW \times RW$. This is basically a classic urn experiment. The anthers produce pollen, R and W each with probability $\frac{1}{2}$. The pistil makes ovules in the same proportions.

The pollen and ovules combine into offspring, zygotes. They do so in the proportions $\frac{1}{4}$ RR, which follows from Popper's multiplication theorem Eq. (P3). $\frac{1}{2}$ $RW = \frac{1}{4}$ $RW + \frac{1}{4}$ WR, by Popper's addition theorem Eq (P2), since RW and WR are genotypically and phenotypically identical. Finally we get $\frac{1}{4}$ WW, again by Eq (P3). Now, we mentioned dominance above. The R allele is dominant over W so, again by Popper's addition theorem we get $\frac{3}{4} = \frac{1}{4} + \frac{1}{2}$ red pea flowers and $\frac{1}{4}$ in the offspring from $RW \times RW$.

The dominance of red over white is similar to the kind of propensity for individual events like the case with two dice, one loaded and the other fair, where it is impossible to separate one die from the other. Just as it is impossible to tell the difference between RW and RR, one phenotype produced by two genotypes. This crossing is illustrated in Figure 4.

The formation of germ cells through meiosis is not always simple and peaceful as they are in red and white peas in Fra Gregor's Monastery Garden. All over the living world one can find disturbances from Mendel's law of equal segregation that are due to phenomena called *segregation distortion* or *meiotic drive*. An allele A_d can be capable of pushing for itself when germ cells are formed. That is, individuals with genotype $A_d A_x$ produce more gametes with A_d than with A_x. This can be caused by purely cellular mechanisms, but there are also known examples where sperms or eggs (in males and females, respectively) actually attack and kill each other in 'The Battle for Transmission' (Lindholm *et al.*, 2016).

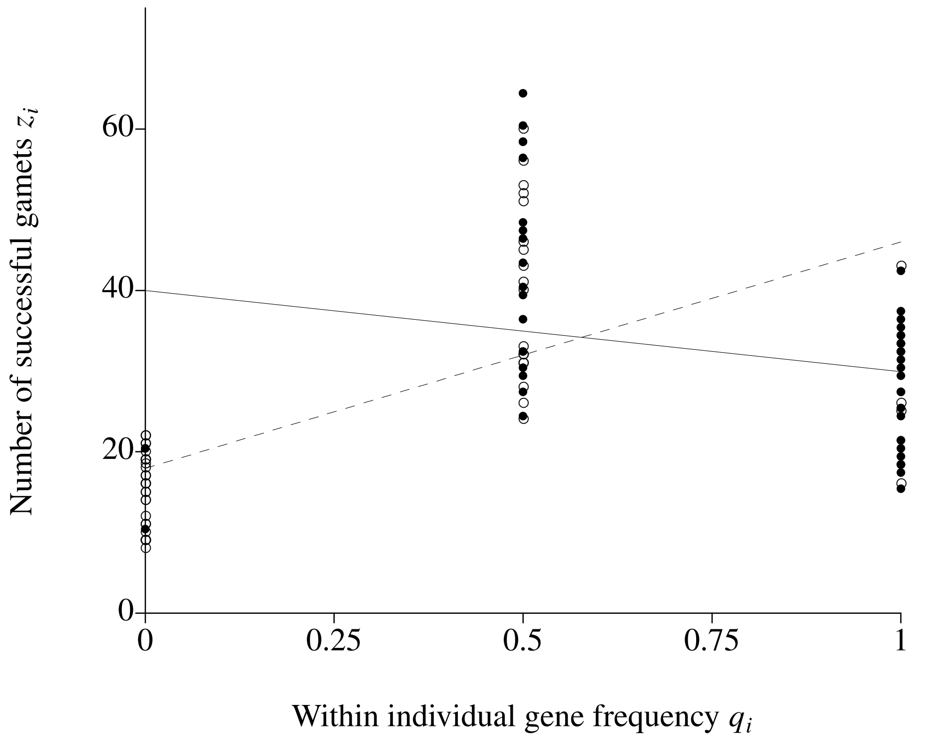

Figure 5. Two hypothetical regression analyses where the number of successful gamets for each individual in the population of 50 individuals assigned to the genotypes *aa*, *Aa* and *AA* randomly but in approximately Hardy-Weinberg proportions (see main text). It is assumed that also assumed that there is an advantage for the heterozygotes (*Aa*) such they have the highest reproductive success (their mean values are assigned 15, 45 and 30 with some random variation for *aa*, *Aa* and *AA*, respectively). The open rings and dashed line represent the situation where the $\overline{q} = 0.25$ and the *A* will increase the next year. The bullets and solid line shows the situation where the $\overline{q} = 0.75$ and the *A* will therefore decrease the following year.

iv

Singular events, such as the births and deaths of individuals can make it difficult to directly perceive any probabilities that are hidden in a plethora of other circumstances. George R. Price[6] modelled selection as a truly mechanistic process. In particular his covariance formulation of the mathematical theory of Natural Selection is regarded as conceptually elucidating (Price, 1970). I will derive Price's equation, and in parallel use Popper's theorems to derive a dynamical system which make it possible to calculate the change of gene frequencies over time.

We assume a single locus with two alleles, *A* and *a*, and calculate the change in frequency of the gene *A* from one generation to the next, ΔQ . In this case his equation looks like this:

$$\Delta Q = \frac{\mathrm{cov}(z, q)}{\overline{z}} \tag{1}$$

Here z_i is the number of successful gametes from individual i ($i = 1 \ldots N$ where N is the population size) and q_i is the frequency of *A* in that individual, $q_i = 0, 1/2, 1$. $\overline{z}$ and $\overline{q}$ are the arithmetic means of the number of successful gametes and frequencies of *A* over all the *N*

[6] Price was American population geneticist and before that he had a career as a physical chemist and as science journalist. He worked in the Manhattan project with the physical and chemical properties of Uranium 235, contracted thyroid cancer during the 1950s. He died by his own hand 1975, possibly induced by hypothyroidism which is both a physical and psychiatric condition following thyroid hormone deficiency. He lived among homeless in London and gave away all his possessions, while working on the evolution of altruism at the Galton Laboratory, University College, London (Harman, 2011).

individuals in the population, respectively. Note that Price adds an alternative formulation

$$\Delta Q = \frac{\beta_{zq}\sigma_q^2}{\overline{z}} \tag{1a}$$

Here β_{zq} is the slope of a regression of z_i, the succesful games on q_i, the frequency of gen A in individual i. Here we can see how natural selection operates using standard statistical analysis. I give a hypothetical example in Figure 5.

Price considers two populations P_1 and P_2, where the former represents the parents and the latter their offspring. We first consider the P_1. Let g_i be the dose of the gene A in individual i. $g = 0, 1, 2$ for a diploid species. Further, let n_z the zygotic ploidy of the species. For a diploid species $n_z = 2$. The frequency of the gene A in P_1 is

$$Q_1 = \frac{\sum\limits_{i=1}^{i=N} g_i}{n_z N} = \frac{\sum\limits_{i=1}^{i=N} n_z q_i}{n_z N} = \overline{q} \tag{2}$$

where we take the sums over all members of population P_1.

In traditional population genetics you do this differently. Here you assume (i) Mendel's law of inheritance and (ii) random mating. (Price assumes nothing of this kind when deriving his equation.) Imagine that we pick up each individual and take a note of its genotype, if it is AA, Aa or aa (basically this is what Price does). From those data we can easily calculate the gene frequency. Now we use the traditional assumptions and Equation (2) to calculate $\overline{q}$. We knew the frequency Q_1 of A in the population P_1. Let us now form the population P_1 as zygotes from the gametes from their parents. This is an urn experiment. We draw two A with probability Q_1^2, it follows from the multiplication theorem, Equation (P3). Likewise, we pick up one A followed by a a with the probability $Q_1(1 - Q_1)$ and one a followed by a A with probability $(1 - Q_1)Q_1$—both using the multiplication theorem Equation (P3) and then we sum them together per Equation (P2). Since the two are identical, sum them together and get $2Q_1(1 - Q_1)$.

$$Q_1 = \frac{2Q_1^2 N + 2Q_1(1 - Q_1)N}{2N} = \overline{q} \tag{2'}$$

Here we have used 2 for g_i for all those that are homozygous and 1 for those that are heterozygous. This way to go from gene frequencies to genotype ones is called Hardy-Weinberg's law:

$$\begin{aligned} AA&: \quad Q^2 \\ Aa&: \quad 2Q(1 - Q) \\ aa&: \quad (1 - Q)^2 \end{aligned}$$

There is nothing more than Popper's theorems (or the corresponding traditional ones), behind this exercise.

Now, turn to the frequency of A in the offspring population P_2 (the next generation). That frequency is determined by the number of successful gametes z_i from each individual i, but also the number of A genes g'_i. We also have to take into account the gametic ploidy n_G. For a diploid species the ploidy is $n_z = 2$ (see above) and consequently the gametic is $n_G = 1$. The frequency of A in individual gametes produced by individual i is then defined by

$$q'_i = \begin{cases} g'_i / z_i n_G & \text{if } z_i \neq 0 \\ q_i & \text{if } z_i = 0 \end{cases} \tag{3}$$

Let $\Delta q_i = q'_i - q_i$ for short. It is the difference between the frequency of A genes per successful gamete from individual i and the frequency of that gene in that individual. This quantity is very important, and I will return to its significance below.

Also, recall that the population covariance between two stochastic variables, lets call them z and q, are defined as

$$\mathrm{cov(z,q)} = \frac{\sum_{i=1}^{N}(z_i - \overline{z})(q_i - \overline{q})}{N} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} \tag{4}$$

Let Q_2 be the frequency of A in the offspring population P_2. Then

$$Q_2 = \frac{\sum_{i=1}^{N} g'_i}{\sum_{i=1}^{N} z_i n_G} = \frac{\sum_{i=1}^{N} z_i n_G q'_i}{\sum_{i=1}^{N} z_i n_G} = \frac{\sum_{i=1}^{N} z_i q'_i}{N\overline{z}} \tag{5a}$$

$$= \frac{\sum_{i=1}^{N} z_i q_i}{N\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} = \overline{z}\,\overline{q} + \frac{\mathrm{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{5b}$$

$$= \overline{q} + \frac{\mathrm{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{5c}$$

Now we are approaching the end of the derivation. Subtract Equation (2) from (5c)

$$\Delta Q = Q_2 - Q_1 = \frac{\mathrm{cov}(z,q)}{\overline{z}} + \frac{\sum_{i=1}^{N} z_i \Delta q_i}{N\overline{z}} \tag{6}$$

All this was to derive Equation (1) from first principles. The second term in Equation (6) is not a bug but a feature. It is the mean value of the differences between the frequency of A genes per successful gamete from individual i and the frequency of that gene in that individual. This is a gross simplification; we assume that all individuals with the same genotype are exactly the same in all other aspects. We assume that all those other aspects somehow even out. The genotypes have become what von Mises refer to as *classes* or *collectives* (von Mises, 1957).

Another factor that will make it non-zero is genetic drift, also known as sampling error. There are also a range of other factors, like sexual selection (Fisher, 1930, p. 146–156). Selection can also operate through meiotic drive and segregation distortion, when a gene is trying to get an upper hand in the Battle of Transmission (Lindholm *et al.*, 2016) rather than in improving survival and reproduction in general. That term is an argument for the propensity interpretation of probability.

The traditional population genetics use another formulation. We start with a variation of Equation (4)

$$\mathrm{cov(z,q)} = \frac{\sum_{i=1}^{N} z_i q_i}{N} - \overline{z}\,\overline{q} = \frac{NQ_1^2 z_{AA} + N2Q_1(1-Q_1)z_{Aa}\,\frac{1}{2}}{N} - \overline{z}\,\overline{q} \tag{4'}$$

where we assume Hardy-Weinberg equilibrium and remember the definition of q_i above. Also we replace the index i on the z_i (the number of successful gametes from individual i) with their genotypes. I also replace Q_1 and $\overline{q}$ with the letter p (they are equal under the given assumptions)

$$\mathrm{cov(z,q)} = p^2 z_{AA} + p(1-p)z_{Aa} - \overline{z}\,\overline{q}$$

Substituting this into Equation (1) (using the p rather than Q) we get

$$\Delta p = \frac{p^2 z_{AA} + p(1-p) z_{Aa}}{\overline{z}} - p \tag{1'}$$

which is derived from the Price equation rather than the conventional way, as Okasha (2024) does it.

V

The Price Equation (1) and its traditional formulation in Equation (1') has lead us to a dynamical system[7] that models the change in gene frequencies between generations. Dynamical systems are typically sets of difference or differential equations, one equation per variable under study. These variables are usually referred to as state variables. A dynamical system can be solved analytically or numerically to yield time series of the states. The solutions are typically of three kinds. (i) Convergence to a point, where state variables move along trajectories towards a point often referred to as an attractor. (ii) An attractor can be a limit cycle when the trajectories, or orbits, returns over and over again to the same or similar states. Finally (iii) it can be chaotic.

The trajectories of a dynamical system have one important feature. They are uniquely determined by their initial conditions, they can diverge from, or converge, to one another, but never intersect. The Equation (1') is of simple kind. Its dynamics converges to one of three equilibrium points.

The change predicted by dynamical systems are in reality affected by various kinds of stochasticities. There will be three kinds of such effects. We have two that we may refer to as *internal,* because they do actually belong to the dynamical systems itself: (i) There could be sampling errors in the mating process causing genetic drift (Crow and Kimura, 1970, p. 367). The genetic drift is dependent on population size and the strength of the other processes in the population. If selection is weak and the population density is low, the outcome could be more or less random. (ii) Demographic stochasticity, basically there might be accidents affecting the birth and death processes (Pielou, 1977). The internal stochasticities are superimposed on selection pressures and effects like sexual selection, meiotic drive etc mentioned above. In addition we have (iii) *external* stochasticity caused by the environment.

Mathematically, the environmental impact is usually Gaussian (normally distributed) disturbances (May, 1974, p. 109–139). Concretely it involves everything from harsh weather to asteroids and the Cretaceous–Paleogene extinction event (Schulte, 2010).

There is an interesting convergence between developments of state variables, orbits or trajectories in dynamical systems, and possible worlds semantics in metaphysics. The total state space can be seen as a universe, and in each possible world the system has its own trajectory (Williamson, 2018; Williamson, 2016). Williamson shows that there is an important class of modalities emerging in the theory of dynamical systems.

The trajectories in a dynamical system are deterministic in exactly the sense Lewis foresaw

> By determinism I do not mean any thesis of universal causation, or universal predictability-in-principle, but rather this: the prevailing laws of nature are such that there do not exist any two possible worlds which are exactly alike up to some time, which differ thereafter, and in which those laws are never violated (Lewis, 1973).

Two trajectories can diverge, but they cannot be ‘exactly alike’ and then all of a sudden differ from a certain time on. Environmental stochasticity would do the job, but that would effectively preclude them from being ‘exactly alike’ for some period. Lewis aims at providing an analysis of causation under this kind of determinism. He notes (in his footnote 7) that

[7] A branch of mathematics, working with difference and differential equations to model processes. I learned it from Luenberger (1979).

Anscombe (1971) proposes that undetermined events should possibly be caused.

Anscombe notes that this is the case for infectious diseases.

> If, then, I have had one and only one contact with someone suffering from such a disease, and I get it myself, we suppose I got it from him. But what if, having had the contact, I ask a doctor whether I will get the disease? He will usually only be able to say, 'I don't know—maybe you will, maybe not' (Anscombe, 1971).

Obviously this is undetermined. However, different diseases have different propensities for spreading. For instance, assume that Anscombe had not contracted any childhood diseases when young. Now, when a parent, she was sitting by the bed when her children had Measles, Chickenpox and Mumps. These infections have basic reproductive rates R_0, 11–15, 7–12 and 11–14, respectively (Keeling and Grenfell, 2000). Obviously, Anscombe's doctor would modify the judgement as to the likelihood that she contracted the disease from that contact event, based on the R_0 value. However the actual cause would still be undetermined. Note that the definition of R_0 is often given as the number of secondary infections caused by a single infected individual in an otherwise healthy population. When $R_0 < 1$ the population has *herd immunity* (Anderson and May, 1985). Again we have a situation where the probabilities of singular events are essential for the understanding of an entire class of phenomena.

The invasibility is actually an extremely useful property in the study of ecological or evolutionary processes. There are a large number of processes where we have no idea on how, or even if, a certain trait is heritable, but we know for sure that there are modifier genes that may cause minor changes and that we can study the early fate of such mutants. This enables to model how evolution would mould such traits. This is the idea behind the theory of Evolutionarily Stable Strategies (ESS) developed for that purpose.

> A 'strategy' is a behavioural phenotype; i.e., it is a specification of what an individual will do in any situation in which it may find itself. An ESS is a strategy such that, if all members of a population adopt it, then no mutant strategy could invade the population under the influence of natural selection (Maynard Smith, 1982).

Here we see again how the properties of few individuals are having crucial importance for the fate of the many. Or in terms of a dynamic system; is the equilibrium state of the system invasible or susceptible to change? Here is where dynamical systems theory meets modalities such as possibility and necessity.

vi

There are a few issues related to numbers and mathematics. The first one is well described by David Lewis

> Another well-known problem for simple frequentism: if spacetime is finite and chance systems get only just so small, then all frequencies are rational numbers. Yet the great majority of real numbers are irrational, and we have no pre-philosophical reason to doubt that chances in a finite world can take irrational values (Lewis, 1994).

This is even more relevant for populations. Ten billion human carriers of mutations is countable, and so is the number of genes. Hence, the number of individuals of any species on earth is natural, it is member of $\mathbb{N}$, therefore frequencies of genes and genotypes have to be rational, they must $\in \mathbb{Q}$. The mathematics we use for dynamical systems are real numbers, and Karl Popper's theorems are based on his expression (reiterated here):

$$p(a, b) = r, \text{where } r \in \mathbb{R}$$

The $P(a|b)$ in Equation (1') is also member of $\mathbb{R}$. We naturally accept the real numbers, as probabilities, gene frequencies, population densities and so forth. This attitude comes easily as long as we accept the platonic view of mathematics. Then mathematical objects do not have real world counterparts requiring ontological commitments (Horsten, 2026).

However, if the members of $\mathbb{R}$ exist and their properties should have meaning for objects in the physical or biological world, then this raises problems. What does it even mean that variables would have values that would take an infinite amount of time to calculate (if we were to calculate them with all their infinite decimals)? This is one of the reasons intuitionistic mathematics does not accept them as such objects (Gisin, 2021).

vii

A second aspect of how Popper presents his theory is that he sees the frequency interpretation as an example how we get knowledge through induction, whereas his propensity theory is forward looking where a setup is evaluated in an experiment. This mirrors his falsification theory (Popper, 1957a).

The frequency theory is looking from some point backwards along a lineage. What worked in the past might be seen as inductive knowledge. The propensity theory is looking forward from the same point. The future is always an experiment, a test that can falsify that knowledge in a changing environment. I have the impression that Karl Popper (1957a) sees his theory this way, that his view of probabilities should be valid at a metaphysical level (Popper, 1978). The propensities as forces are, at a conceptual level, essential for the understanding of all that we see as changes out in nature. Dynamicity is at the core of evolution.

The past and induction can explain the present. However, to use them to predict the future is what Popper (1957b) refers to as historicism. This was the reason he initially rejected the theory of Natural Selection. He did eventually endorse it again 1978 when he was invited to give the first Darwin lecture at Darwin College, Cambridge. He proposed that

> Ecological conditions like those that favour the evolution of open behavioural programmes sometimes also favour the evolution of the beginnings of consciousness, by favouring conscious choices (Popper, 1978).

By 'open behavioural programmes' he refers to behaviours that are not in detail coded for by the DNA, but were moulded by learning. He suggested that individuals had probabilities or propensities for acquiring them, that had evolved through natural selection.

Organisms are individuals and each of them has unique properties while propagating genetic information to future generations. Just for that reason we have to flip the coin and look upon how individual properties manifest themselves at the level of the population. For instance: An individual has a gene, a population has a gene frequency. Individuals mate, give birth, survive and finally die. Populations grow, decline and go extinct.

Natural selection works on the individual level, hence the probabilities that matter for Evolution out there in Nature must be for singular events. Furthermore, Natural selection is not looking at your CV and past merits. It looks at your next achievements. Do you survive and reproduce the next year? Natural selection is not induction, it is falsification.

Acknowledgements: This note is dedicated to my grandchildren. I thank Gertrud for her understanding with my absentmindedness when I was thinking of probabilities.

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